

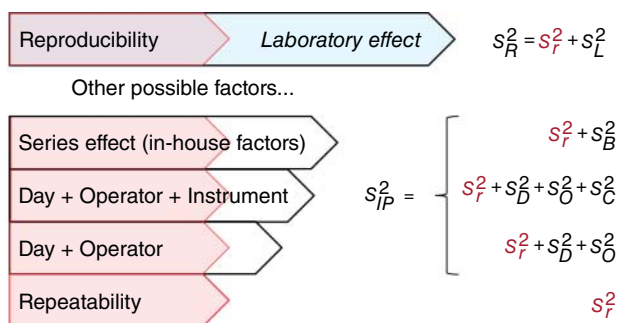


Figure 3.1 Graphical representation of diverse total variance decomposition, affording diverse sources of variation. Repeatability is always present and represented by a red arrow.

- When replicates are performed in the same laboratory, but varying one or more operating conditions (also called factors), such as day, reagent batches, operator, instrument settings, or any other factor, the precision is estimated under *intermediate precision condition*, thus-called because it is in between repeatability (the smallest variation) and reproducibility (largest variation).
- When replicates are performed in different laboratories on the same sample with supposedly the same operating procedure, the precision is estimated under *reproducibility condition*. This means that different test portions of the working sample are analyzed by different operators on different days using almost identical standard operating procedures but adapted within different laboratories.

Though this book deals with in-house validation, the interlaboratory approach to precision is interesting because concepts and computation techniques are transposable from interlaboratory to in-house validation. For statisticians, the parameter used to estimate data dispersion is the variance or its square root, the standard deviation. Therefore, when speaking about precision, it means speaking about variances or standard deviations. Figure 3.1 is an attempt to illustrate the complexification of the precision variance from the repeatability condition, where only random effect plays a role, to the reproducibility condition, where sources of variation are numerous.

3.1.2 Role of Series for Data Collection

The term “series” was introduced in Chapter 2 and appears several times in all following chapters. It is critical and extensively used because it is a basic element of laboratory management and in-house validation. It can be defined as a set of measurements, repeated or not, run under repeatability condition. A series puts together all the measurements performed under the same conditions, for example, same method, same laboratory, same day, same operator, same calibration curve, etc. It is the basic planning of laboratory work that consists of grouping samples intended to be analyzed using the same analytical method. The main goal of the management of samples in series is to reduce the fixed costs of the method, as explained in Section 4.4.

When replicates are performed in a single laboratory and planned in different series, we are under intermediate precision

variation are introduced, such as days (D), operators (O), instrument settings I, and reagents. There is a full spectrum of possible combinations and possible intermediate precision standard deviations. Intermediate precision condition is recommended (or required) for in-house validation and measurement uncertainty (MU) estimation, as explained in Section 6.5.

Figure 3.1 illustrates an important idea that is developed in ISO 5725 standards. Precision variance can be decomposed into several components, each characterizing the effect of a factor of variation and a global replication condition. Finally, precision is quantified by different variances, which are combined. Depending on the organization of measurements, the intermediate precision variance s_{IP}^2 may contain a variable number of components.

Repeatability variance (or within-series)

$$s_r^2$$

Between-series variance

$$s_B^2$$

Intermediate precision variance

$$s_{IP}^2 = s_r^2 + s_B^2 \quad (3.1)$$

Between-laboratories variance (or laboratory effect)

$$s_L^2$$

Reproducibility variance

$$s_R^2 = s_r^2 + s_L^2 \quad (3.2)$$

An example of an interlaboratory study is used to illustrate how precision parameters can be computed and interpreted. It consists of a panel of 11 laboratories that were requested to make three replicate measurements of Pb on a sample of homogenized freeze-dried mussel tissue. The resulting dataset, called LEAD, is also used in Section 7.4 to illustrate a possible procedure for evaluating MU from the results of an interlaboratory analysis.

This dataset is described in Table 3.1. Strictly speaking, the definition of the reproducibility condition is not exactly satisfied because participants used their own analytical technique. Results for each participant are collected in Table 3.2.

Table 3.1 LEAD – description of the dataset.

Title of the dataset	LEAD
Reference	[7]
Number of laboratories	11 participants ($I = 11$)
Replicates/laboratory	($J = 3$)
Measurand	Lead concentration in mollusk tissue, in mg/kg
Methods	ICP-MS spectrophotometry

Table 3.2 LEAD – original dataset.

Laboratories	Replicates (mg/kg)			Average
	1	2	3	
Lab 01	2.08	2.00	2.01	2.03
Lab 02	2.00	1.93	1.89	1.94
Lab 03	2.10	2.44	1.96	2.17
Lab 04	2.45	2.34	2.49	2.43
Lab 05	1.95	1.89	1.93	1.92
Lab 06	1.85	1.91	1.89	1.88
Lab 07	2.01	2.00	2.06	2.02
Lab 08	2.00	2.09	1.98	2.02
Lab 09	2.11	2.03	2.14	2.09
Lab 10	2.02	1.98	1.97	1.99
Lab 11	2.02	2.00	2.04	2.02

Precision parameters calculated following ISO 5725-2 recommendations are:

Parameters	Symbol	Value
Number of laboratories	I	11
Number of replicates	J	3
Grand mean	$\bar{\bar{Z}}$	2.047
Repeatability variance	s_r^2	0.00751
Interlaboratory variance	s_L^2	0.01945
Reproducibility variance	s_R^2	0.02696
Repeatability std. dev.	s_r	0.0867
Reproducibility std. dev.	s_R	0.1642

As stated before, the scope of ISO 5725 standards was to define common rules for comparing results issued from several laboratories and solving eventual trade disagreements. Therefore, the maximum acceptable difference between two measurements from two different laboratories is a classic acceptance criterion. It is called the Reproducibility Limit and conventionally denoted R . It is derived from the standard deviation of reproducibility and is equal to:

$$R = 2.83 \times s_R$$

The coefficient 2.83 is obtained by considering the 95% confidence interval of the difference between two measurements having the same variance s_R^2 . The origin of this coefficient is linked to the variance of the difference between two random

variables with the same variance. The limit of reproducibility is obtained using the following computational rules:

Difference between lab A and lab B

$$D = |Z_A - Z_B|$$

Variance of the difference

$$s_D^2 = 2 \times s_R^2$$

Standard deviation of the difference

$$s_D = \sqrt{2} \times s_R$$

Limit of reproducibility: 95% confidence interval of the difference

$$2.0 \times s_D = 2.0 \times \sqrt{2} s_R = 2.83 \times s_R$$

The coefficient 2.0 used in the last formula is called the “coverage factor.” This is standardized and explained in Section 6.2. This gives, for the LEAD example:

$$R = 2.83 \times 0.1642 \simeq 0.465 \text{ mg/kg}$$

A decision rule can be derived from this criterion as follows: if two lead measurements performed in two different laboratories on a sample around 2 mg/kg differ by less than 0.465 mg/kg, they are considered equivalent; both laboratories give the same result. Figure 3.2 illustrates the LEAD dataset. Measurements are spread around the grand mean, and some results from laboratories 3 and 4 are not highly consistent with the results from the others; several of them do not comply with the limit of reproducibility. The question of abnormal results will be addressed in Section 3.4.2 about outliers.

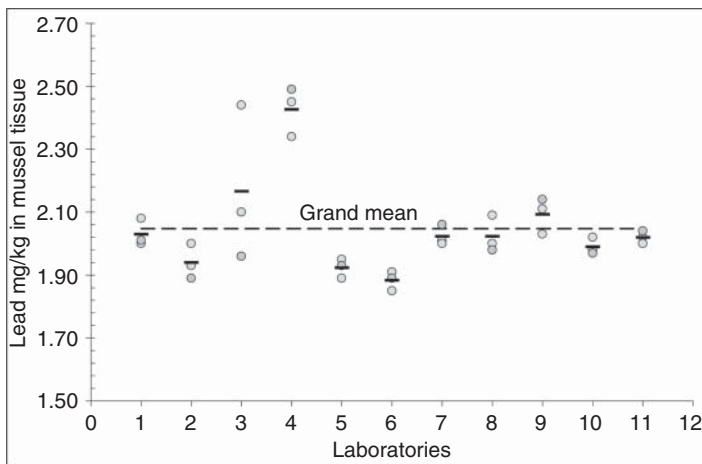


Figure 3.2 LEAD – illustration of interlaboratory study. The grand mean is a dotted line, small horizontal bars are laboratory means.

3.2 Analysis of Variance (ANOVA)

3.2.1 Computation of Precision Parameters

The idea behind interlaboratory data interpretation is that each measurement value is a combination of a common repeatability precision that characterizes the method and the effect of the laboratory bias. In an interlaboratory study, the effect of the laboratory factor is the influence of each participant on the determination of the contents X of the material sent to participants. The corresponding model is:

$$Z_{ij} = X + L_i + E_{ij}$$

- Z_{ij} is the measurement value of replicate j with $(1 \leq j \leq J)$ for laboratory i with $(1 \leq i \leq I)$.
- X is the “true value” of the study material (a discussion about this concept is proposed in Section 4.1.2). It is constant.
- L_i is the effect of the i th laboratory on the global result. It is not constant and may take random values.
- E_{ij} is the random residual error of each replicate.

Therefore, total precision, i.e. total variance of Z_{ij} must be decomposed into two main components: the laboratory effects and the residual errors. The classic statistical tool used to decompose the variance is the analysis of variance (ANOVA). It was first described in the 1920s by Fisher. Its principle is to distribute the total variance of a dataset into several pieces, each corresponding to a source of variation. This operation is possible if the dataset has a predefined structure.

For instance, the LEAD dataset structure is simple: each data is a replicate, hierarchically linked to one laboratory. The laboratory entity is called a *factor*. Many types of ANOVA may exist, depending on the structure of the dataset and the number of factors. According to the nomenclature adopted by statisticians, the ANOVA used for interpreting interlaboratory study is a *one-way random effects ANOVA* as opposed to a *one-way fixed effects ANOVA*.

The effects L_i of the laboratory factor are said to be random because they are not controlled by the organizer, who randomly selects a group of laboratories among all existing ones [8, 9]. On the contrary, fixed factor levels are controlled, like temperature, volume, or pH. Random and fixed factors can be mixed into more complex designs that require specialized software to be processed. An example of a more complex variance decomposition model is available in [10].

Any participating laboratory must perform replicates because the goal is to measure variances. For each laboratory, it is essential that replicates be performed under repeatability conditions, i.e. within the same series. Measurement replication is no longer required if the laboratory takes part in proficiency testing. This is another type of collaborative study devoted to estimating trueness but not precision. The statistical modeling of proficiency testing is different, as will be explained in Section 7.4.

In the context of in-house validation, the factor used to do the ANOVA is no longer the laboratory but the series, which often corresponds to the date when the assays